

Benzene is an important organic chemical compound with chemical formula C_6H_6 . Its molecule is composed of 6 Carbon atoms and 6 Hydrogen atoms. Its chemical structure can be described as a hexagonal ring with alternating double bonds. In benzene there are 6 sigma bonds between carbon atoms and 6 sigma bonds between carbon and hydrogen atoms. It has also 3 pi bonds between carbon atoms. Therefore 12 sigma and 3 pi bonds are present in benzene molecule.

23. Numbers of sigma and pi bonds in benzene are:

- (a) 3, 3 (b) 3, 6
(c) 12, 3 (d) 12, 6
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

See the explanation of above question.

B. Alcohol

Notes

- Alcohol is an organic compound with a hydroxyl ($-OH$) functional group on an aliphatic carbon atom. Because $-OH$ is the functional group of all alcohols, we often represent alcohols by the general formula $R-OH$ where R is an alkyl group.
- In other words, the hydroxyl derivatives of hydrocarbons are termed as alcohols.
- Molecular formula and name of some alcohols are as follows :
 - (i) Methyl Alcohol (Methanol) – CH_3OH
 - (ii) Ethyl Alcohol (Ethanol) – C_2H_5OH
 - (iii) n-Propyl Alcohol – $CH_3CH_2CH_2OH$
 - (iv) Isopropyl Alcohol – $(CH_3)_2CHOH$
 - (v) n-Butyl Alcohol – $CH_3(CH_2)_2CH_2OH$
 - (vi) Isobutyl Alcohol – $(CH_3)_2CHCH_2OH$

Methyl Alcohol (Methanol or Carbinol) :

- Its chemical formula is CH_3OH .
- It acquired the name **wood alcohol** or **wood naphtha** because it was once produced primarily by the destructive distillation of wood.
- Today it is mainly produced by hydrogenation of carbon monoxide.
- It is the simplest alcohol, consisting of a methyl group linked to a hydroxyl group.
- It is light, volatile, colourless, flammable liquid with a distinctive odour similar to that of ethanol (Ethyl alcohol).
- At room temperature, it is a **polar liquid**.
- Its boiling point is $64.7^\circ C$ ($148.5^\circ F$) and it solidifies at $-93.9^\circ C$ ($-137^\circ F$).

- It forms **explosive mixtures** with air and burns with a nonluminous flame. After burning in the air or in oxygen it forms carbon dioxide and water.
- It is a very toxic substance. Methanol ingestion is an uncommon form of poisoning that can cause severe metabolic disturbances, blindness, permanent neurologic dysfunction and death.
- It increases the **acidity** of the body after ingestion.
- Due to its toxic property, it is used to denature the ethyl alcohol which is not suitable for drinking.
- It is soluble in water and some other organic solvents.
- A crystalline compound $CaCl_2 \cdot 4CH_3OH$ is formed by the reaction of methyl alcohol and anhydrous calcium chloride, so it can not be dried up by anhydrous calcium chloride.

Uses of Methyl Alcohol :

- It is used as a solvent for paint, celluloid, shellac, etc.
- The mixture of petrol and methyl alcohol is used as fuel in automobiles.
- It is used in the preparation of dyes, perfumes, medicines etc.
- It is used in the preparation of formaldehyde, formic acid, acetic acid, methyl chloride, methyl salicylate, etc.
- It is used in the preparation of the **methylated spirit**. A mixture of methyl alcohol and ethyl alcohol is known as methylated spirit or **denaturated** spirit, which is not suitable for drinking.
- The mixture of methyl alcohol (40%) and water (60%) is used in radiators of automobiles as **antifreeze**.

Ethyl Alcohol (Ethanol) :

- Ethyl alcohol (C_2H_5OH) is commonly called **alcohol**.
- It is found in forms of ethyl esters of carboxylic acids in fruits, plants and fragrant oils.
- It is the main component of all types of wines, so it is called the **spirit of wines**.
- It is made from the **fermentation** or **chemical breakdown** of sugars by yeast, so it is also called **grain alcohol**.

Absolute Alcohol–

- Absolute alcohol is 100% ethanol.
- The **rectified spirit** (95.6% ethanol + 4.4% water) is a constant boiling point mixture (Azeotropic solution) and water cannot be removed from it by simple fractional distillation method.
- DNA is insoluble in alcohol so we use absolute alcohol for precipitation of a good amount of DNA.
- It is extremely hygroscopic, transparent, colourless, volatile liquid used as a solvent and injected into nerves and ganglia for relief of pain.

Power Alcohol -

- Power alcohol is a mixture of petrol, ethanol and minute amount of aromatic compound such as benzene. It is used as motor fuel.

Properties of Ethyl Alcohol :

- It is colourless and inflammable liquid.
- Its boiling point is 78.1°C and melting point is -114.1°C , which allows it to be used in antifreeze products.
- It has a special odour (alcoholic odour).
- Its taste is sharp.
- It is soluble in water and many organic solvents.
- It is a neutral compound. It does not affect the pH of water. It is neutral to acid-base indicators.
- It forms a crystalline solid $\text{CaCl}_2 \cdot 3\text{C}_2\text{H}_5\text{OH}$ with anhydrous calcium chloride, so ethyl alcohol cannot be dried up by anhydrous calcium chloride.

Uses of Ethyl Alcohol :

- Ethanol is present in alcoholic beverages such as beer, wine, whisky, etc. So, it is also called drinking alcohol.
- Ethanol is used as **antiseptic** for sterilizing wounds.
- Ethanol is used in cough syrups, digestive syrups and tonics.
- Ethanol is being mixed with petrol and is used as motor fuel. This mixture is **called power alcohol**.
- A mixture of ethanol and water has a lower freezing point than water. This mixture is known as **antifreeze** and is used in radiators of vehicles in cold countries and at hill stations.
- Ethanol is used for the preparation of chloroform, iodoform, etc.
- Ethanol is used as a hypnotic (induces sleep).
- Ethanol is used in the preparation of methylated spirit.
- Ethanol is used to preserve the biological specimens.
- Ethanol is used in spirit level and alcohol thermometer.

Question Bank

1. Ethyl alcohol is made unfit for drinking by the addition of the following :

- Potassium Cyanide
- Methanol and Pyridine
- Acetic acid and Pyridine
- Naphthalene

R.A.S./R.T.S.(Pre) 1999

Ans. (b)

Ethyl alcohol becomes poisonous by the addition of methanol and pyridine.

2. The offending substance in the liquor tragedies leading to blindness etc. is –

- Ethyl alcohol
- Amyl alcohol
- Benzyl alcohol
- Methyl alcohol

I.A.S. (Pre) 1996

Ans. (d)

Methyl alcohol is a colourless liquid. It is soluble in water. Its smell is like a pungent, unpleasant and intensive flavour. In each ratio, it is poisonous while in excess, the intake of methyl alcohol results in madness, blindness and its excess consumption can also cause death.

3. Which one of the following is also known as 'wood spirit'?

- Methyl alcohol
- Ethyl alcohol
- Ethylene glycol
- Glycerol

U.P.P.C.S. (Mains) 2015

Ans. (a)

Methyl alcohol (CH_3OH) is also called wood spirit. It is obtained by the destructive distillation of wood.

4. Fermentation of sugar leads to –

- Ethyl alcohol
- Methyl alcohol
- Acetic acid
- Chlorophyll

47th B.P.S.C. (Pre) 2005

Ans. (a)

Ethyl alcohol is formed by the fermentation of sugar, which is made of glucose and fructose.

5. Glucose is converted to ethyl alcohol by the enzyme

- maltase
- invertase
- zymase
- diastase
- None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

Glucose is converted to ethyl alcohol (ethanol) by the enzyme zymase. Zymase catalyzes the fermentation of sugar (glucose and fructose) into ethanol and carbon dioxide. This enzyme complex naturally occurs in yeast and other anaerobic organisms. This enzyme is used in the preparation of ethanol and alcoholic beverages commercially. Invertase enzyme converts sucrose (cane sugar) into glucose and fructose. Maltase enzyme converts maltose into glucose while diastase enzyme converts starch into maltose.

6. To whom molasses are excellent raw material for production?

- Glycerin
- Urea